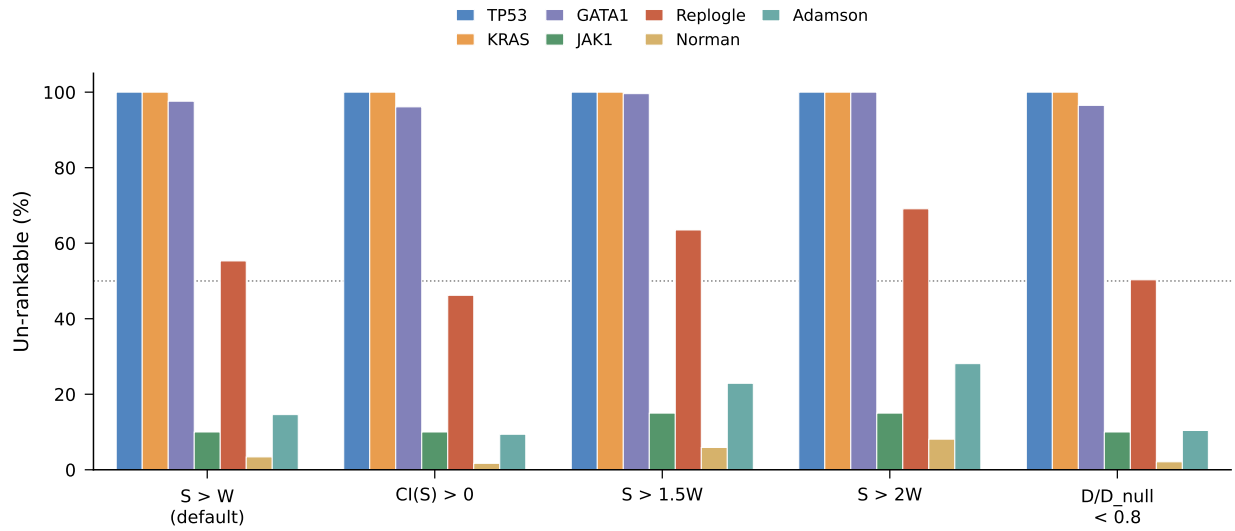


Supplementary Information

Measurement and model limits of allele-specific single-cell perturbation prediction

This document contains the full parameter tables underlying the variant feature representation, the prediction methods and the split-half rankability analysis. Complete assignments and code are also available at <https://github.com/Boom5426/AllelePerturb>.



Supplementary Figure 1: **The rankability verdict is robust to the choice of criterion.** Fraction of un-rankable perturbations for the seven datasets (the four AllelePerturb genes and three gene-level Perturb-seq atlases) under five rankability criteria: the primary signal-versus-noise rule $S > W$ (default); a bootstrap lower-bound rule ($CI(S) > 0$, the lower 95% confidence bound of S above zero); two increased-stringency rules ($S > 1.5W$ and $S > 2W$); and a ratio threshold, under which a variant is un-rankable when $D_{\text{self}}/D_{\text{null}} \geq 0.8$. Absolute fractions shift with stringency, but the ordering across datasets is preserved under every criterion: TP53 and KRAS remain $\sim 100\%$ un-rankable, GATA1 stays high, JAK1 stays low ($\sim 10\%$), and among gene-level atlases Replogle is consistently the highest and Norman the lowest. Source values are listed in `results/rankability_sensitivity.csv`.

Supplementary Table 1: **Variant feature (θ) structural and hotspot annotations.** Per-gene residue assignments for the three structural components of the biophysical feature vector. The fold-core term f_v was additionally set to 0.5 for annotated interaction regions and 0.0 elsewhere. The hotspot/pathogenic marker h_v was defined exclusively from external prior knowledge, with no reference to the single-cell outcome.

Gene	Structural core ($f_v = 1$)	Catalytic switch ($s_v = 1$)	/	Hotspot / pathogenic ($h_v = 1$)
TP53	DNA-binding domain, residues 94–293	–		COSMIC/IARC missense hotspot codons 175, 176, 179, 220, 238, 245, 248, 249, 273, 282, 285
KRAS	G-domain, residues 1–169	G12, G13, Q61		COSMIC/IARC missense hotspot codons 12, 13, 59, 61, 117, 146
GATA1	Zinc fingers (204–228) and ZF1 ZF2 (258–282)	Zinc-finger cysteines		Zinc-coordinating cysteines 204, 207, 225, 228, 258, 261, 279, 282; ClinVar pathogenic / likely-pathogenic annotations
JAK1	Kinase domain, residues 866–1154	–		Functional-module membership: JH2 pseudokinase (583–855), JH1 kinase (875–1153)

Note. An earlier version of h_v used the top quintile of measured effect size, which leaks the evaluation outcome into the feature; it was removed and the entire evaluation grid was re-run, with ranking performance unchanged (θ -only PDS shifted by at most 0.02 and every method remained at chance).

Supplementary Table 2: **Implementation and sampling parameters.** *Regression heads (top):* the six heads were crossed with three feature spaces (θ , ESM, ESM+ θ) to form the 18 in-house feature-model combinations; all heads used a fixed random seed and were fit by minimizing squared error on the training variants. *Split-half and power-analysis sampling grid (bottom):* parameters for the energy-distance split-half analysis (Fig. 4) and the depth power analysis (Figs. 4, 5), each configuration summarized by the median and the 2.5–97.5 percentile interval across seeds.

Head	Family	Settings
Ridge regression	Linear (L2)	$\alpha = 1.0$
Lasso regression	Linear (L1)	$\alpha = 0.01$; up to 2,000 iterations
Random forest	Tree ensemble	50 trees; maximum depth 6
Gradient-boosted trees	Tree ensemble	50 estimators; maximum depth 3; multi-output wrapper
k -nearest-neighbors	Neighbor	$k = 3$
Multilayer perceptron	Neural network	hidden layers 128 and 64; ReLU; up to 500 iterations

Sampling parameter	Values
Split-half subsample sizes (per half)	25, 50, 100, 150, 250, 400 cells
Representation spaces	50-component PCA (canonical), gene space, pathway-score space
Random seeds per configuration	50
Power-analysis depths (per variant)	50, 100, 150, 300 cells
Minimum cells for inclusion	50 (split-half); 5 (pseudobulk profiles)

Note. Linear and neighbor heads regress directly onto the full expression profile. Because tree ensembles scale poorly to thousands of output genes, for output spaces larger than 200 genes the random forest and gradient-boosted heads were fit on a q -component PCA of the training profiles and predictions were mapped back to gene space by the inverse transform, with $q = \min(50, G, n_{\text{train}} - 1)$.

Supplementary Table 3: **Detection, identification and prediction: a decision table for allele-specific single-cell perturbation prediction.** Three nested questions of increasing difficulty locate a dataset by its failure mode: detection (a variant versus wild type; assay sensitivity), identification (one variant versus another; ground-truth resolution) and prediction (a held-out variant from its features; computational generalization). Prediction is only meaningful once identification is achievable. Detection is graded rather than binary: in the three datasets with low measured effect size the variant-to-wild-type separation is marginal (near the replicate-noise floor; Fig. 4c,e), so ‘marginal’ detection denotes a weak but non-zero separation from wild type. Marginal identification likewise denotes a small fraction of resolvable sibling pairs with no reliable recovery of a variant’s nearest sibling. Because each gene here is also a distinct dataset, assay and cell context, a row locates a measurement and modelling regime rather than an intrinsic property of the gene.

Detection	Identification	Prediction	Limiting regime	Dataset in this study
no	–	–	assay-limited	(none)
marginal	no	not evaluable	measurement-limited	TP53, KRAS
marginal	marginal	not evaluable	measurement-limited	GATA1
yes	yes	no	model-limited	JAK1
yes	yes	yes	successful regime	(target)

Supplementary Table 4: **Published perturbation models under a unified allele-level evaluation.** Perturbation-discrimination score (PDS, cosine) with 95% bootstrap confidence intervals, pooled over five generalization splits and four genes; chance is 0.50. Variant-conditionable models are scored directly; gene-keyed models are allele-blind by construction and reported by resolution coverage. Every method was scored through one harness (identical held-out variants, pseudobulk profiles, tie-aware PDS and permutation null).

Model family	Method	PDS (95% CI)	Pearson- δ
<i>Variant-conditionable (run to completion)</i>			
Optimal-transport flow	CellFlow	0.51 (0.48–0.54)	0.12
Attribute disentangle- ment	Biolord	0.49 (0.47–0.52)	0.13
Latent arithmetic (dose)	scVIDR	0.50 (0.48–0.53)	0.13
Latent arithmetic	scGen	0.49 (0.46–0.52)	0.18
Conditional-flow gener- ative	PerturbNet [†]	0.54 (0.51–0.57)	0.13
Feature regression	baselines (Fig. 2)	0.49–0.52	0.55–0.65
<i>References</i>			
Gene identity	Gene-mean	0.49 (0.46–0.52)	0.66
No perturbation	WT-null	0.50	–
<i>Allele-blind by construction (resolution coverage 4/470 = 0.85%)</i>			
Foundation model	scGPT	n/a [‡]	–
Gene knowledge graph	GEARS	n/a [‡]	–
Virtual-cell (one-hot)	STATE	n/a [‡]	–
<i>Did not converge</i>			
Compositional autoen- coder	variant-CPA	non-finite loss	–

[†]PerturbNet generates within a 50-dimensional principal-component subspace in which random predictions already score 0.52; full-space-at-chance predictors (Ridge-esm, Gene-mean) projected into the same subspace are likewise inflated to 0.52, and its exceedance over a subspace permutation null is not significant (permutation $P = 0.13$). [‡]These models condition on perturbed-gene identity and assign an identical prediction to every variant of a gene, so an allele-level PDS is undefined (a category error).

Supplementary Table 5: **Per-gene detection ratio, achievable ceiling and model performance.** Split-half energy-distance ratio $D_{\text{self}}/D_{\text{null}}$; oracle discrimination ceiling (independent second measurement, estimating attainable discrimination); fraction of allele pairs resolvable above replicate noise; fraction of variants rankable at native depth; and best in-house model PDS (canonical multi-seed harness). Chance is 0.50.

Gene	$D_{\text{self}}/D_{\text{null}}$	Oracle ceiling (95% CI)	Pairs resolvable	Rankable (native)	Best model PDS [†]
TP53	0.96	0.485 (0.445–0.525)	0%	0%	0.51
KRAS	1.00	0.500 (0.440–0.566)	0%	0%	0.54
GATA1	0.88	0.572 (0.549–0.595)	3.8%	2.4%	0.50
JAK1	0.21	0.792 (0.742–0.838)	78.5%	90%	0.52

[†]Best model is the highest-scoring in-house feature-regression head per gene under the canonical multi-seed harness (published variant-conditionable models are compared separately in Supplementary Table 4). PerturbNet is excluded; its higher per-gene values (0.53–0.54) arise from its 50-dimensional principal-component subspace, in which random predictions already score 0.52.

Supplementary Note: robustness and sensitivity controls.

Statistical independence of variants. Because some residues carry more than one substitution, the 470 variants are not fully independent. Under the random split, 47% of held-out variants share a residue with a training variant (positional 0%, mechanistic 10%, low-depth 37%, compatibility 54%); the random split is therefore an optimistic setting, yet allele discrimination remained at chance under it and under the residue-disjoint positional split alike. Resampling residue-level clusters rather than individual variants in the bootstrap left the per-method PDS intervals essentially unchanged: all 23 non-null predictors continued to overlap the 0.50 chance level (only PerturbNet excluded it, the subspace artifact described above), and Pearson- δ remained above zero for every feature-model head. The direction-versus-ranking dissociation is therefore not an artifact of treating correlated substitutions as independent units. Source: results/reviewer_controls/residue_independence_summary.csv.

Wild-type reference construction. To test whether the detection floor reflects the variant effect or the heterogeneity of the wild-type reference, we compared, at matched depth, the variant replicate distance D_{self} , the variant-to-wild-type distance D_{null} and the wild-type-to-wild-type distance D_{wtwt} from an independent split of the wild-type cells (energy distances, 50-component PCA space). For TP53, KRAS and GATA1 the three distances coincided ($D_{\text{wtwt}}/D_{\text{null}} = 0.91\text{--}1.09$ across 50–300 cells per half), so a variant differs from wild type no more than wild type differs from itself, and D_{self} , which involves no wild-type cells, sits at the same level: the floor is set by the absence of variant signal relative to sampling noise, not by an inflated or heterogeneous control. JAK1 shows the contrast, with D_{null} eight- to thirty-fold above D_{wtwt} ($D_{\text{wtwt}}/D_{\text{null}} = 0.03\text{--}0.12$). The ratios are stable across wild-type sampling depth, so the floor is not a depth artifact. Source: results/reviewer_controls/wt_control_diagnostic.csv.

Metric-family invariance. The detection floor is not specific to energy distance. The split-half window ratio $D_{\text{self}}/D_{\text{null}}$ (50-component PCA, 50 cells per half) stayed near one for TP53, KRAS and GATA1 and low for JAK1 under all five distance families tested (energy, RBF maximum-mean-discrepancy, sliced-Wasserstein, cosine and Euclidean): TP53 1.01–1.03, KRAS 0.95–0.99, GATA1 0.90–1.04, JAK1 0.05–0.40. Source: results/canonical/metric_family_floor.csv.

Gene-shared versus allele-specific decomposition. Splitting each effect as $\delta_v = \bar{\delta}_g + r_v$ apportions the measured response between a component shared by all variants of a gene and an allele-specific

residual; the shared fraction is a property of each dataset rather than of the metric, at 72% (TP53), 59% (KRAS), 29% (GATA1) and 51% (JAK1). What predictors actually recover follows instead from scoring the predictions against the residual: across all 24 predictors the residual Pearson- δ lay between -0.08 and 0.03 , so conventional direction recovery is carried almost entirely by the gene-shared component. An oracle scored on the allele-specific residual profiles stayed at chance for TP53 and KRAS (0.480 and 0.478), was marginal for GATA1 (0.588) and high for JAK1 (0.892), so the allele-specific residual is reproducible only where the measurement resolves sibling variants.

Representation robustness. Under identical splits, regression heads and scoring (single deterministic pseudobulk draw; the multi-seed convention used in the main text gives PDS 0.49 – 0.52), every variant representation left allele discrimination at chance while recovering direction similarly: PDS-cosine 0.458 – 0.473 and Pearson- δ 0.628 – 0.659 for the biophysical θ vector, ESM-1v, four ESM2 constructions (global, mutant-minus-wild-type global, mutated-residue and local-window) and AlphaMissense; a featureless training-gene-mean predictor reached Pearson- δ 0.66 at PDS 0.44 . The scoring procedure is identical across representations, but variant coverage is not, because each representation is defined only where its input exists: AlphaMissense scores were available for 374 of the 470 benchmark variants (84 of 98 for TP53, 78 of 92 for KRAS, 201 of 254 for GATA1 and 11 of 26 for JAK1), giving 389 scored variant-by-split instances against 518 for θ , 511 for ESM-1v and 497 for the ESM2 constructions. Correlations between AlphaMissense pathogenicity and measured effect size are therefore reported per gene only (Spearman -0.08 for TP53, -0.05 for KRAS and 0.09 for GATA1); cross-gene pooling is not interpretable here, because both the pathogenicity distribution and the scale of the measured transcriptional effect differ between genes. Sources: `results/reviewer_controls/rep_grid_summary.csv`, `am_summary.csv` and `am_effect_spearman.csv`.

Oracle ceiling sensitivity. The replicate ceiling was unchanged by depth (300-cell versus full-depth) and stayed at chance for TP53 and KRAS even for a perfect prediction scored against the benchmark’s noisy pseudobulk truth (TP53 0.488 , KRAS 0.490); GATA1 remained marginal (0.569) and JAK1 high. Shrinking a prediction fully to the gene mean lowered discrimination to 0.500 for every gene, so mean-shrinkage denoising does not raise the ceiling.

Train-only preprocessing. Refitting the per-gene standardization on training variants and wild-type cells only (excluding held-out variants) left every gene at or below chance: per-gene PDS-cosine, all-cell versus train-only, was $0.47/0.44$ (TP53), $0.49/0.50$ (KRAS), $0.46/0.46$ (GATA1) and $0.46/0.45$ (JAK1), with direction recovery essentially unchanged (mean Δ PDS = -0.01), so the reported all-cell standardization is conservative with respect to the null. Source: `results/canonical/leakage_control_summary.csv`.